

Topological ~~set~~ set:

Def.: (X, τ) $\tau \subseteq P(X)$

1. $\emptyset, X \subseteq \tau$

2. closed under Arbitrary union

3. closed under Finite intersection

metric space:

$$(X, d) \rightarrow d: X \times X \rightarrow [0, \infty)$$

$$1. d(x, y) = 0 \iff x = y$$

$$2. \text{ symmetry}$$

$$3. d(x, y) \leq d(x, z) + d(z, y)$$

A vector space V : $\mathbb{F}(\mathbb{R}, \mathbb{C})$

addition: $V \times V \rightarrow V$

scalar mult $\times$: $\mathbb{F} \times V \rightarrow V$

$1v = v$ (unit)

$0v = 0$ (zero)

norm space: $(V, \|\cdot\|)$

$$\|\cdot\|: V \rightarrow [0, \infty)$$

- $\|v\|=0 \Leftrightarrow v=0$
- $\|\alpha v\| = |\alpha| \|v\|$
- $\|u+v\| \leq \|u\| + \|v\|$

Inner product space: $(V, \langle \cdot, \cdot \rangle)$

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$$

$$\|v\| = \sqrt{\langle v, v \rangle}$$

$$\|x\|_p = \sqrt[p]{\sum x_i^p} \rightarrow p\text{-norm}$$

$$\text{quadratic norm: } p \in S_{++}^n \quad |x|_p = x^T p x$$

$$\forall x: \alpha \|x\|_a \leq \|x\|_b \leq \beta \|x\|_a$$

نرم معادل:

operator norm

$$\|X\|_{a,b} = \sup_u \{ \|Xu\|_a / \|u\|_b \leq 1 \} = \sup_u \frac{\|Xu\|_a}{\|u\|_b}$$

$$a=b=2 \rightarrow \|X\|_{2,2} = \sup \frac{\|Xu\|_2}{\|u\|_2} = \sigma_{\max}(X)$$

$$a=b=1 \rightarrow \|X\|_{1,1} = \max_j \sum_i |x_{ij}|$$

dual norm:

$$\|Z\|_{\star} = \sup_x \{Z^T x \mid \|x\| \leq 1\}$$

$$|Z^T x| \leq \|x\| \|Z\|_{\star}$$

for p norms:

$$\|x\|_{p,\star} = \|x\|_q$$

$$\frac{1}{p} + \frac{1}{q} = 1$$

$$\|Z\|_{2,\star} = \sup_x \underbrace{\{Z^T x \mid \|x\|_2 \leq 1\}}_{\langle \cdot, \cdot \rangle} = \|Z\|_2$$

$$\|X\|_2 = \sigma_{\max}(X)$$

$$\|X\|_{2,\star} = \sum \sigma_i(X) = \text{nuclear norm}$$

$$\langle A, X \rangle = \text{tr}(A^T X) \leq \sum_i \sigma_i(A) \sigma_i(X) \leq \sum \sigma_i(A)$$

$$\sigma_i(X) \leq \|X\|_2 \leq 1$$

$$A = U \Sigma V^T$$

$$X = U \Sigma^T$$

$$\langle A, x \rangle = \text{tr}(A^T U v^T) = \text{tr}\left(v^T \underbrace{U^T U}_I v\right) = \text{tr}(v^T v) = \sum \sigma_i$$

Eigen value:

$$A x = \lambda x$$

$$A = U \Lambda U^{-1}$$

$$\text{f.o.r } A \in S^n : \quad A = Q \Lambda Q^T, \quad Q^T Q = I$$

$$\det(A) = \prod \lambda_i$$

$$\text{trace}(A) = \sum \lambda_i$$

$$\lambda_{\max}(A) = \sup \frac{x^T A x}{x^T x}$$

$$\forall x: \lambda_{\min}(A) \leq \frac{x^T A x}{x^T x} \leq \lambda_{\max}(A) \quad A \in S^n$$

Proof: $A = Q^T \Lambda Q$

$$y = Q x \xrightarrow{Q^T Q = I} \|y\|_2 = \|x\|_2$$

$$x^T A x = \underbrace{x^T Q^T}_{y^T} \Lambda \underbrace{Q x}_y = y^T \Lambda y = \sum \lambda_i y_i^2 \leq \lambda_{\max} \frac{\sum y_i^2}{\sum x_i^2} \leq \lambda_{\max} \frac{x^T x}{x^T x}$$

معین بودن A (PD)

1) $\forall x \neq 0 \quad x^T A x > 0$

2) $\forall i \quad \lambda_i(A) > 0$

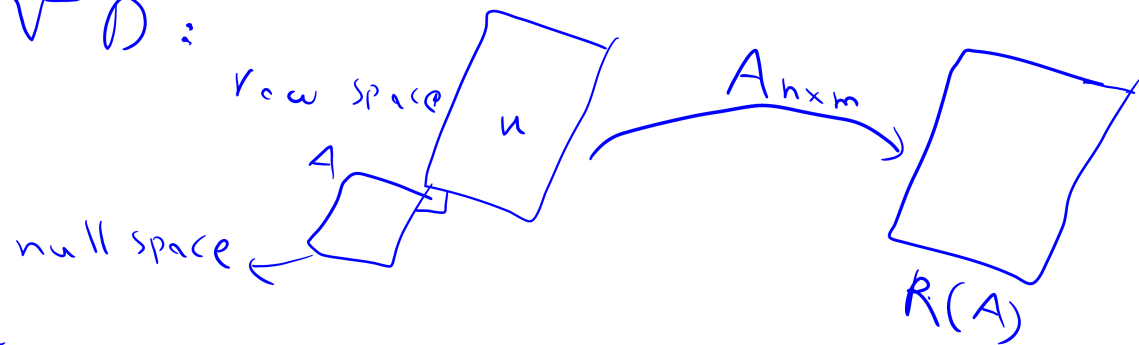
3) Sylvester test

$$\begin{bmatrix} \text{ } \\ \text{ } \\ \text{ } \end{bmatrix}$$

4) $\forall i \quad \text{Pivot}_i(A) > 0$

5) $A = C C^T \quad \text{col } C \text{ independent}$

SVD:



$$A \underbrace{[u_1, u_2, \dots, u_n]}_{\text{unitary}} = \underbrace{[v_1, v_2, \dots, v_m]}_{\text{unitary}} \Sigma$$

$$U^T U = I$$

$$V^T V = I$$

$\text{rank}(A)$

$$A = \sum_{i=1}^{\text{rank}(A)} \sigma_i v_i v_i^T$$

$$A U = V \Sigma \rightarrow A = V \Sigma U^T$$

$$\sigma_{\max}(A) = \frac{x^T A y}{\|x\|_2 \|y\|_2} = \sqrt{\lambda_{\max}(A^T A)} = \sqrt{\lambda_{\max}(A A^T)}$$

Pseudo-inverse:

$$A^+ = V \Sigma^{-1} U^T = \lim_{\varepsilon \rightarrow 0} (A^T A + \varepsilon I)^{-1} A^T = \lim_{\varepsilon \rightarrow 0} A^T (A A^T + \varepsilon I)^{-1}$$

$A_{n \times m}$:

$$\text{rank} = n \quad A^+ = (A^T A)^{-1} A^T$$

$$\text{rank} = m \quad A^+ = A^T (A A^T)^{-1}$$

$$\text{rank} = n = m \quad A^+ = A^{-1}$$

$$A x = b \longrightarrow A^T A x = A^T b \longrightarrow x = (A^T A)^{-1} A^T b$$

$$f(x) = \frac{x^T A x}{x^T x} \quad A \in S^n$$

$$\left. \begin{array}{l} \text{S.t. } x^T A x \\ \text{S.t. } x^T x = 1 \end{array} \right\} \rightarrow L = x^T A x - \lambda (x^T x - 1)$$

$$\nabla L = 2Ax - 2\lambda x = 0$$

$$\rightarrow \underline{Ax = \lambda x}$$